

Arjun Prasad

Senior Backend Engineer | Distributed Systems & Payments Infrastructure
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SUMMARY

Backend engineer with 4.5+ years building high-throughput payment systems at PhonePe (500M+ users). Currently on UPI Core Transactional team handling 20,000+ RPS and 500M+ daily transactions. Previously built greenfield microservices processing 4M+ transactions/day across BBPS infrastructure. Created a platform-wide async framework cutting infra costs 45%. Led Active-Active DC migration with circular replication. Strong focus on reliability, scalability, and end-to-end production ownership.

SKILLS

Java (8-17) | Dropwizard, Spring Boot, Hibernate, JUnit | MariaDB (Sharded), Aerospike | RabbitMQ, MemQ, Event-driven Architecture | Microservices, Active-Active DC, Docker, CI/CD, Nginx | System Design, Distributed Systems, Database Sharding, REST APIs, DR/BCP

EXPERIENCE

PhonePe -- UPI Core Transactional Team | SDE-2 (Backend)

Mar 2026 - Present | Pune | 20,000+ RPS, 500M+ transactions/day

- Operating on India's highest-traffic payment rail (UPI) -- mission-critical uptime, 20K+ RPS peak traffic
- Building E2C refund status tracking for 2-2.5L daily refunds -- enabling in-app visibility where users currently have zero tracking (must check bank statements)
- Owning NPCI contract architecture, UDIR (UPI Dispute Resolution) flows, and external PSP integrations (GPAY, Paytm)

PhonePe -- BBPS Pod | Software Engineer to SDE-2 (Backend)

Feb 2022 - Feb 2026 | Pune | 500 RPS peak, 4M+ billpay transactions/day

UPMS -- Greenfield Microservice (0 to 1)

- Independently designed and shipped a new microservice -- architecture, sharded DB, proxy layer, recon, biller kill-switches -- zero to 500+ RPS in 5 months
- Replaced polling-based bill fetching with push-based presentment, driving 20% transaction growth; completed NPCI certification and monolith data migration

Axon -- Async Messaging Framework (45% Infra Cost Reduction)

- Built a reusable async messaging framework from scratch abstracting RMQ/MemQ with unified publish API, fallback strategies, and dynamic worker buckets
- Adopted org-wide -- reduced per-service integration from days to minutes; 45% infrastructure cost reduction across PhonePe

Active-Active DC Migration

- Led migration from Active-Passive to Active-Active: redesigned DB schema for circular replication, migrated async workers to MemQ (DC-aware)
- Achieved zero-downtime DC failover; blast radius reduced from 100% to ~50%

Status Check Pipeline Overhaul

- Reduced stuck transactions from 20,000+/day to <100/day; automated resolution from 9+ hours to <10 minutes; eliminated wrong refund risk entirely

Other Key Contributions

- Piped Gas Meter Reading: Built end-to-end in 1 week (contract to production), demonstrated at GFF 2025. PhonePe-exclusive integration.
- Paylink migration to modern framework -- +15% monthly transactions, zero downtime. Loan API 1.1 -- led NPCI spec compliance independently.
- Launched 6 bill categories; created SOP eliminating engineering dependency. Mentoring 3 engineers, 117 code reviews, 6 DR drills conducted.

Samsung Research Institute -- Software Engineer

Jul 2021 - Feb 2022 | Noida

- Reduced audio device setup time by 30% through Bluetooth module optimization (1,000+ devices)
- Samsung SWC Professional certification (Data Structures & Algorithms)

EDUCATION

B.Tech Computer Science & Engineering

National Institute of Technology, Delhi (2017-2021)

HIGHLIGHTS

- Received "Distinguished" rating (highest tier) at PhonePe -- "Completed the implementation flawlessly and quickly" -- Engineering Manager
- Scale progression: 500 RPS (BBPS) to 20,000+ RPS (UPI Core) -- consistently trusted with increasing responsibility
- Built 2 greenfield systems (UPMS service + Axon framework), both adopted in production at scale
- 345+ merge requests, 605 engineering tasks delivered, 356 architecture/design documents authored across 4+ years